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Prepregnancy Care and Counseling
A Review

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IMPORTANCE Prepregnancy care and counseling optimize maternal health before conception to improve outcomes for mothers and infants. In the US, 66.4% of reproductive-aged women have at least 1 modifiable risk factor for adverse pregnancy outcomes.

OBSERVATIONS For all individuals desiring pregnancy, recommended interventions include folic acid supplementation; cessation of tobacco, alcohol, cannabis, and opioids; immunizations against hepatitis B virus, varicella, and rubella; and screening for syphilis and HIV. Folic acid use before pregnancy is associated with reduced fetal neural tube defects (relative risk [RR], 0.67; 95% CI, 0.52-0.87). Maternal tobacco smoking is associated with increased risks of stillbirth (summary RR [sRR], 1.46; 95% CI, 1.38-1.54), neonatal death (sRR, 1.22; 95% CI, 1.14-1.30), and perinatal death (sRR, 1.33; 95% CI, 1.25-1.41). Screening for and treatment of syphilis and HIV prior to and during pregnancy decrease rates of fetal and neonatal infection. Prepregnancy immunizations against hepatitis B virus, varicella, and rubella decrease neonatal infection and mortality. Individuals using tobacco, alcohol, cannabis, and opioids should receive counseling and treatment prior to pregnancy (eg, buprenorphine or methadone for opioid use disorder). For individuals with chronic disease, routine health examinations and contraceptive care in the year before conception can optimize pregnancy timing and are associated with decreased risk of severe maternal morbidity. Compared with planned pregnancies, unintended pregnancies are associated with increased risk of postpartum depression (15.7% vs 9.6%; adjusted odds ratio [aOR], 1.51; 95% CI, 1.40-1.70), preterm birth (9.4% vs 7.7%; aOR, 1.21; 95% CI, 1.12-1.31), and low infant birth weight (7.3% vs 5.2%; aOR, 1.09; 95% CI, 1.02-1.21). Weight loss prior to conception is recommended for individuals with a body mass index of 25 or greater because overweight and obesity are associated with increased risk of gestational diabetes, gestational hypertension, and cesarean delivery. Among patients with pregestational diabetes (type 1 or 2), hemoglobin A_{1c} of less than 6.5% is associated with a decreased risk of fetal anomaly compared with hemoglobin A_{1c} of 6.5% or greater. Cardiovascular complications such as hypertension and heart failure occur in 15% of pregnancies and are more common among those with preexisting cardiovascular disease. These patients should receive counseling on maternal and neonatal risk, monitoring, and medication management by specialists in cardiology and maternal fetal medicine.

CONCLUSIONS AND RELEVANCE Prepregnancy counseling and care reduce maternal morbidity and neonatal morbidity and mortality. Primary care–based discussion of reproductive goals, immunizations, screening for infections and substance use, and risk-reducing interventions such as folate supplementation can optimize outcomes in individuals contemplating pregnancy.

Multimedia

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Prepregnancy care and counseling encompass a range of interventions to improve maternal and neonatal outcomes. While optimization of health and preventive care benefits all reproductive-aged people, regardless of pregnancy intention, primary care–based discussion of reproductive goals and provision of individualized education, screening, and risk-reducing interventions can support the increased motivation to optimize health in individuals contemplating pregnancy (**Figure**).[1,2]

Worldwide, 1.9 billion women are eligible for reproductive health care.[3] In 2023, global maternal mortality, defined by the World Health Organization (WHO) as death during pregnancy, during childbirth, or within 42 days of delivery, was 197 per 100 000 live births (260 000 deaths),[4] compared with 18.6 deaths per 100 000 live births in the US (669 deaths).[5] Medical conditions such as hypertension, diabetes, and infections (eg, HIV, COVID-19) are the second leading cause of maternal death.[6] In the US, 36% of reproductive-aged women report at least 1 chronic medical condition and 66.4% have a modifiable risk factor for congenital birth defects.[7,8]

Conditions and exposures with relevance to prepregnancy care are commonly encountered in primary care (**Table 1**). Prepregnancy education on healthy behaviors is associated with a 17% lower risk of neonatal mortality and a 39% increase in antenatal care seeking.[1] Guidelines from the American College of Obstetricians & Gynecologists (ACOG), American Academy of Family Physicians (AAFP), and WHO highlight modifiable risk factors, targeted education, and interventions (**Table 2**).[19-21] This Review summarizes the current evidence for prepregnancy care and counseling with a primary care focus.

Methods

PubMed, Scopus, and the Cochrane databases were searched with the aid of a medical librarian for English-language studies on prepregnancy care and counseling (January 1, 2015–December 31, 2025). A separate search for articles was performed focusing on the role of primary care clinicians in prepregnancy care, for which no date range was used. Key words used for the PubMed searches are listed in eAppendix 1 in the Supplement. The primary care–focused search, which was limited to systematic reviews, meta-analyses, clinical trials, cohort studies, and case-control studies, yielded 210 articles. The general prepregnancy care PubMed search yielded 448 articles. Reference lists of the included studies were searched for additional relevant studies. This Review includes 118 articles, consisting of 2 randomized clinical trials, 9 longitudinal prospective observational studies, 11 review articles, 15 systematic reviews and meta-analyses, 42 practice guidelines, 33 cross-sectional studies, and 6 systematic reviews. Relative effect measures are presented when absolute risk data are not available.

Discussion

The **Box** provides a list of commonly asked questions on prepregnancy care and counseling. eAppendix 2 in the Supplement provides resources for prepregnancy care and counseling and related topics and eAppendix 3 in the Supplement lists the Healthy People 2030 prepregnancy care–related objectives.

Reproductive Life Planning

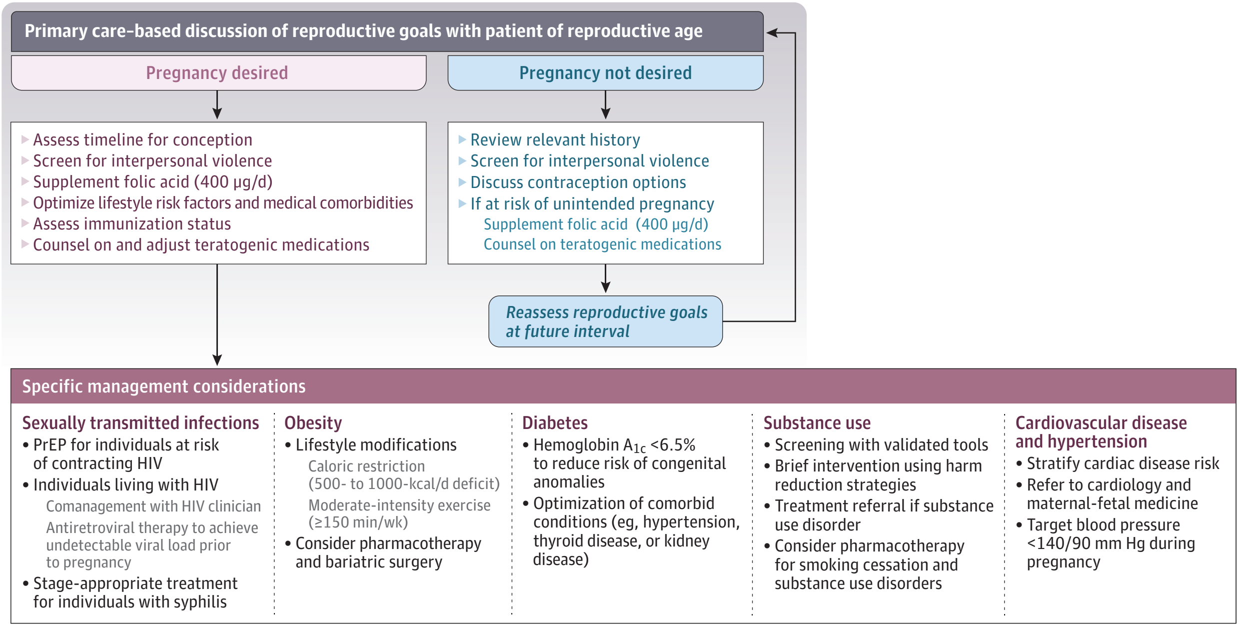
Unintended pregnancies are associated with adverse maternal and neonatal outcomes. A systematic review and meta-analysis of 36 studies (14 cohort studies and 22 cross-sectional studies, of which 12 studies used population-based data sources; 524 522 participants) reported that compared with individuals with planned pregnancies, those with unintended pregnancies had higher prevalence of postpartum depression (15.7% vs 9.6%; adjusted odds ratio [aOR], 1.51; 95% CI, 1.40-1.70; 10 studies; 82 673 patients), preterm birth (9.4% vs 7.7%; aOR, 1.21; 95% CI, 1.12-1.31; 10 studies; 94 351 patients), and low infant birth weight (7.3% vs 5.2%; aOR, 1.09; 95% CI, 1.02-1.21; 8 studies; 87 547 patients).[22] In a 2022 analysis of Medicaid claims examining prepregnancy care (1 514 759 births), receipt of contraceptive services in the year before conception was associated with 8% reduced odds of severe maternal morbidity from conditions such as heart failure, eclampsia, and sepsis (aOR, 0.92; 95% CI, 0.88-0.95), and routine physical or gynecologic examinations resulted in lower odds of severe maternal morbidity for those with chronic disease (aOR, 0.79; 95% CI, 0.71-0.88).[2]

Reproductive life planning helps individuals set and achieve goals about if, when, and how to have children[23] and can occur at annual primary care wellness visits.[24] This care includes contraceptive counseling for those not currently attempting to conceive; for individuals planning pregnancy, information should be provided about return to fertility after discontinuing contraception, prepregnancy health optimization, and avoidance of short interpregnancy intervals, defined as less than 6 months between birth and conception.[25] A systematic review of 129 observational studies (46 874 843 pregnancies) reported that compared with an interpregnancy interval of 18 to 23 months, an interpregnancy interval of less than 6 months was associated with increased risk of preterm birth (pooled OR, 1.55; 95% CI, 1.47-1.63), low birth weight (pooled OR, 1.42; 95% CI, 1.31-1.55), fetal death (pooled OR, 1.52; 95% CI, 1.36-1.70), and birth defects (pooled OR, 1.12; 95% CI, 1.04-1.22).[26] Reproductive counseling should also educate patients about age-related decline in fertility and pregnancy risks with advanced maternal age (≥35 years), such as pregnancy loss or fetal aneuploidy.[27,28]

Return of fertility occurs within 2 to 4 menstrual cycles after discontinuation of contraceptives, except for depot medroxyprogesterone acetate, which takes 5 to 8 menstrual cycles.[29] For patients experiencing difficulty with conception, ACOG recommends infertility evaluation after 1 year of attempted conception for women younger than 35 years, evaluation after 6 months for those aged 35 to 39 years, and immediate evaluation for those aged 40 years or older. Women with risk factors for infertility, such as painful or irregular periods (which may be caused by undiagnosed endometriosis or adenomyosis), history of endometriosis, pelvic inflammatory disease, or a partner with low sperm count, should be evaluated sooner than 6 months.[30] Individuals with medical conditions such as lupus, congenital cardiac disease, or past pregnancy complications such as preterm birth, stillbirth, or preeclampsia should be referred to a maternal-fetal medicine specialist prior to conception.[31]

Pregnancy-associated violence is a leading cause of maternal death.[32] The US Preventive Services Task Force recommends screening for intimate partner violence in all women of reproductive age.[33] Multiple validated intimate partner violence screening tools are available, as provided by the Women's Preventive Services Initiative.

Figure. Flow Diagram of Approach to Prepregnancy Care and Counseling



PrEP indicates preexposure prophylaxis.

Prepregnancy Counseling and Therapies Recommended for All Individuals Desiring Conception

Folic Acid

Fetal neural tube development occurs early in the first trimester, with neural tube defects affecting 1 in 1000 pregnancies.[34] Folate, a B-complex vitamin, plays an essential role in nucleic and amino acid metabolism during neural tube formation. Folic acid supplementation before pregnancy is effective at preventing fetal neural tube defects.[19,20,35,36] A systematic review of 6 observational studies including 12 524 patients reported a relative risk (RR) of 0.67 (95% CI, 0.52-0.87) for neural tube defects with folic acid or multivitamin use compared with nonuse of folic acid.[37] A folate dose of 400 µg/d is recommended for average-risk individuals, and 4 mg/d is advised for high-risk individuals, such as those with a seizure disorder taking anticonvulsant therapy or fetal neural tube defects in a prior pregnancy.[20,34]

Immunizations

Immunization status should be assessed as part of prepregnancy care.[20] Vaccine-preventable infections, including hepatitis B virus, varicella, and rubella, can cause congenital infections during pregnancy, with increased rates of fetal loss with rubella in the first trimester. Influenza is associated with increased maternal morbidity during pregnancy and COVID-19 infection is associated with increased maternal and neonatal morbidity and mortality.[38-40] If not up to date, recommended immunizations for individuals planning pregnancy include tetanus, diphtheria, and pertussis (Tdap)[41]; measles-mumps-rubella (MMR); hepatitis B virus; influenza; COVID-19; and varicella. Avoidance of pregnancy for 1 month following live attenuated vaccination (MMR and varicella) is recommended. Vaccination for pertussis (Tdap) is recommended during every pregnancy to confer passive neonatal immunity. If not previously received during the year, influenza and COVID-19 vaccines are recommended for pregnant individuals during respiratory illness season.[42] Human papillomavirus vaccination prior to conception in reproductive-aged individuals is advised, but human papillomavirus vaccination is not recommended during pregnancy.[20]

Medication Review

ACOG and the AAFP recommend reviewing all medications and supplements as part of prepregnancy care, which allows for discontinuation of medications that may have adverse fetal effects and continuation of low-risk, indicated medications.[19,20] A cross-sectional study of prenatal medication exposure in 2 223 749 US pregnancies (2011-2018; commercial insurance database) reported a prevalence of 2.3% for known teratogenic medication use (eg, sulfamethoxazole-trimethoprim, topiramate, lisinopril, warfarin) and 4.8% for potential teratogenic medication use (eg, gabapentin, methimazole, lithium, meloxicam).[43]

To better contextualize prescription drug risks during pregnancy and lactation, in 2015 the US Food and Drug Administration implemented a revised labeling system to include description of risks, replacing the previous letter categories of A, B, C, D, and X.[44] Medication management for individuals with chronic medical conditions should be individualized with consideration of maternal disease control and fetal exposure, and any necessary dosing adjustments or medication changes should be made prior to pregnancy.[19,20]

Genetic Counseling

Prepregnancy genetic counseling supports informed reproductive decision-making by identifying heritable conditions, assessing risk to offspring, and discussing options such as carrier screening (to determine if parents are carriers of genetic variants for inherited diseases) and preimplantation genetic testing. Couples with autosomal dominant conditions (eg, neurofibromatosis) should be referred

Table 1. Prevalence of Conditions With Relevance to Prepregnancy Care and Counseling

Prepregnancy condition/exposure	Prevalence in US women of reproductive age, % (95% CI)	Pregnancy risks
Overweight	49.8[9]	Gestational diabetes, hypertension, cesarean delivery
Obesity	24.3[9]	Gestational diabetes, hypertension, cesarean delivery
Intimate partner violence (lifetime prevalence among adult women)	47.3 (45.9-48.7)[10]	Maternal morbidity and mortality
Cigarette smoking	20.1 (19.1-21.0)[11]	Preterm birth, low birth weight, perinatal mortality
Hypertension	14.5[12]	Preeclampsia, fetal growth restriction
Alcohol use disorder	12.3 (11.9-13.5)[13]	Fetal alcohol spectrum disorder
Cannabis use	11.8 (11.5-12.0)[14]	Low birth weight, preterm birth, perinatal mortality
Opioid use, nonmedical	10.8 (9.3-12.5)[15]	Neonatal abstinence syndrome, maternal overdose
Diabetes (types 1 and 2)		
Overall	4.7 (3.4-5.9)[16]	Congenital anomalies
Uncontrolled (subset of overall cohort)	51.5 (37.0-65.8)[16]	
Cardiovascular disease	4[17]	Maternal morbidity and mortality
Syphilis	17.7 per 100 000[18]	Congenital syphilis, stillbirth, preterm birth

for genetic counseling. Carrier screening evaluates for autosomal recessive conditions (eg, cystic fibrosis, spinal muscular atrophy, hemoglobinopathy) and X-linked conditions (eg, fragile X, Duchenne muscular dystrophy) to identify higher-risk couples. ACOG recommends that carrier screening be universally offered to individuals considering pregnancy.[20,45] Primary care clinicians may refer patients to genetic counselors or obstetric specialists to receive pretest and posttest counseling.

Prepregnancy Counseling and Therapies for Individuals With Medical Conditions

Obesity

Obesity promotes chronic inflammation, leading to insulin resistance and disruption of the hypothalamic-pituitary-ovarian axis, which can lead to anovulation and infertility.[46] A systematic review and meta-analysis of 83 studies (2 cross-sectional, 2 case-control, and 79 cohort studies; 1 966 026 patients) reported an association between elevated prepregnancy body mass index (BMI; calculated as weight in kilograms divided by height in meters squared) (overweight and obesity) and complications such as gestational diabetes (OR, 2.92; 95% CI, 2.18-2.40), hypertension (OR, 2.08; 95% CI, 1.72-2.53), and cesarean delivery (OR, 1.44; 95% CI, 1.41-1.47).[47] A Canadian population-based cohort of 227 000 pregnancies reported that compared with individuals with a BMI of 36, those with a BMI of 40 had higher risk of preeclampsia (21.4% vs 18%), gestational diabetes (16.9% vs 14.5%), macrosomia (4.3% vs 3.6%), and stillbirth (0.5% vs 0.4%) ($P < .01$ for all comparisons).[48] Larger reductions in BMI among women with obesity (ie, 30% reduction; BMI of 40 vs BMI of 28) were associated with reduction in cesarean delivery (53.5% vs 40.2%), shoulder dystocia (4.2% vs 4%), neonatal intensive care unit admission for 48 hours or longer (5% vs 4.2%), and in-hospital neonatal death (0.6% vs 0.5%) ($P < .01$ for all comparisons).[48] For individuals with obesity, weight loss should be recommended during prepregnancy counseling and care to decrease the risk of these adverse outcomes.[47,49,50]

First-line therapy for weight loss includes caloric restriction (500- to 1000-kcal/d deficit) and exercise (150-300 min/wk of moderate-intensity activity or 75-150 min/wk of vigorous activity with resistance training 2 to 3 times a week).[51,52] In a cohort of 241 Danish women with obesity before undergoing fertility treatment, monthly interviews with a nurse certified in motivational interviewing were associated with weight loss (9.3 kg for intervention vs 7.3 kg for control; $P < .01$).[53] To achieve a BMI of 25 or lower prior to pregnancy, some individuals may also take medications[20] such as glucagon-like peptide 1 (GLP-1) receptor agonists and phentermine-topiramate.[20,52] However, phentermine and topiramate are contraindicated during pregnancy due to teratogenicity; individuals trying to conceive should discontinue these medications at least 2 months prior to conception, based on expert opinion.[54,55] GLP-1 receptor agonists are associated with improved menstrual and ovulation regularity and increased pregnancy rates among women with polycystic ovarian syndrome and obesity.[56] In animal models, GLP-1 receptor agonists have been associated with congenital visceral and skeletal anomalies and decreased fetal weight[57]; however, emerging human data have not demonstrated increased risks of congenital anomalies in pregnancies exposed to GLP-1 receptor agonists.[58] A retrospective cohort study of 149 790 singleton pregnancies in women with obesity reported that compared with individuals who did not take GLP-1 receptor agonists, those who discontinued GLP-1 receptor agonists prior to or early in pregnancy had higher rates of excess gestational weight gain (65% vs 49%; risk ratio, 1.32; 95% CI, 1.19-1.47), preterm delivery (17% vs 13%; risk ratio, 1.34; 95% CI, 1.06-1.69), gestational diabetes (20% vs 15%; risk ratio, 1.30; 95% CI, 1.01-1.68), and hypertensive disorders of pregnancy (46% vs 36%; risk ratio, 1.29; 95% CI, 1.12-1.49).[59] The American Diabetes Association and the Endocrine Society recommend cessation of GLP-1 receptor agonists prior to pregnancy; to date, ACOG has not issued recommendations regarding discontinuation of GLP-1 receptor agonists.[60,61]

Diabetes

Pregestational diabetes (type 1 and 2) affects 1% of pregnancies worldwide[62] and is associated with a 6% to 12% risk of congenital anomalies, most commonly involving the cardiac (eg, ventricular septal defects, conotruncal abnormalities) and central nervous (eg, neural tube defect) systems.[63] Fetal exposure to elevated blood glucose levels affects organ development and alters metabolism, resulting in increased risks of stillbirth, macrosomia, and neonatal hypoglycemia. Among patients with diabetes, pregnancy can

Table 2. Clinical Practice Recommendations for Prepregnancy Care and Counseling From Major Professional Societies

Category	American Academy of Family Physicians (2023)[19]	American College of Obstetricians & Gynecologists (2019)[20]	World Health Organization (2013)[21]
Reproductive life planning	• Inquire whether the patient would like to become pregnant within the next year • Counseling on contraception	• Ask about pregnancy intentions: "Would you like to become pregnant in the next year?" • Discussion of fertility, pregnancy spacing, and contraception	• Provide age-appropriate comprehensive sexuality education, contraceptives • Counseling on awareness and treatment of infertility
Intimate partner violence	• Universal screening for women of reproductive age • Referral and support services for patients who screen positive	• Universal screening during prepregnancy care • If screen positive: assessment of immediate safety, provision of community resources	• Recognize signs of violence • Provide health care services, referral, and psychosocial support
Folic acid supplementation	Folic acid supplementation beginning at least 1 month before conception	Folic acid supplementation to reduce risk of neural tube defects	Folic acid supplementation for women of reproductive age
Immunizations	Update immunizations for patients who desire pregnancy: hepatitis B virus, MMR, Tdap, human papillomavirus, varicella, COVID-19, and influenza	• Assess immunization status annually for Tdap, MMR, hepatitis B virus, and varicella • All patients should receive an annual influenza vaccination	Vaccination against rubella, tetanus, diphtheria, and hepatitis B virus
Medication review	• Review medications • Avoidance of teratogenic medications if potential for pregnancy, reduce essential medications to lowest effective dose	• Review prescription and nonprescription medications and supplements • Adjustment of potentially teratogenic medications in collaboration with prescribing clinician	Not specifically addressed
Genetic counseling	Screen for risk of genetic conditions; offer genetic counseling	Screen for risk of genetic conditions; offer genetic counseling, carrier screening	Screen for risk of genetic conditions; offer genetic counseling, carrier screening
Obesity	• Counseling on effects of 10% weight loss to reduce pregnancy complications • Ideal BMI of 18.5-24.9 prior to pregnancy	Recommend attainment of normal BMI before attempting pregnancy	Nutrition education and counseling (eg, lower caloric intake, increase physical activity, structured weight loss program, continued breastfeeding)
Pregestational diabetes (types 1 and 2)	• Glycemic control (HbA_{1c} <6.5%) • Offer contraception until HbA_{1c} controlled • Medication safety counseling (eg, insulin vs other glucose-lowering medications)	• Glycemic control (HbA_{1c} <6.5%) • Optimal weight management (BMI of 18.5-24.9) • Consider thyroid screening and evaluation for underlying vasculopathy (eg, retinal examination, 24-hour urine protein testing, and electrocardiography)	• Glycemic control • Exercise and nutritional counseling • Screening for preexisting type 2 diabetes every 1-3 years if previous gestational diabetes diagnosis
Hypertension	Counseling on risks of uncontrolled hypertension, blood pressure goals, and pregnancy-safe antihypertensive medications	Assessment of antihypertensive medication safety, consideration of additional testing for end-organ damage in those with long-standing or uncontrolled hypertension (eg, retinopathy, ventricular hypertrophy, kidney disease)	Not specifically addressed
Sexually transmitted infections	• Age- and risk factor-based STI screening • Individuals with childbearing potential with HIV: discuss reproductive goals, provide ART to maintain undetectable viral load before and during pregnancy • PrEP for individuals at high risk	• Age- and risk factor-based STI screening, safe-sex counseling • Women with HIV: comanagement with HIV clinician, provision of ART to attain undetectable viral load before and during pregnancy • PrEP for individuals at high risk	• Age-appropriate comprehensive sexuality education, STI screening, promotion of safe-sex practices • Provision of ART for prevention of maternal-child transmission and PrEP
Substance use and exposures	Screening for alcohol, tobacco, cannabinoids, caffeine, or illicit drug use	Screening for alcohol, nicotine, and other substances, including prescription opioids and other medications used for nonmedical reasons	• Screening for tobacco, alcohol, and other substances • Provision of brief intervention and treatment including pharmacologic and behavioral counseling services

Abbreviations: ART, antiretroviral therapy; BMI, body mass index (calculated as weight in kilograms divided by height in meters squared); HbA_{1c}, hemoglobin A_{1c}; MMR, measles-mumps-rubella; PrEP, preexposure prophylaxis; STI, sexually transmitted infection; Tdap, tetanus, diphtheria, and pertussis.

exacerbate diabetes-related maternal complications such as retinopathy and nephropathy, and increases risk of myocardial infarction and ketoacidosis.[63] In contrast to hemoglobin A_{1c} (HbA_{1c}) targets in nonpregnant adults (HbA_{1c} <7%),[64] target HbA_{1c} should be less than 6.5% for those planning conception.[60] A Canadian cohort study included 3459 births among women with prepregnancy diabetes who had HbA_{1c} measured within 90 days before conception and in early pregnancy to midpregnancy (conception through 21 weeks' completed gestation). Mean HbA_{1c} decreased from 7.2% preconception to 6.4% in early pregnancy to midpregnancy, and a 0.5% lower periconception HbA_{1c} was associated with decreased risk of congenital anomalies (14.4%; absolute risk difference, −0.99%; 95% CI, −1.79% to −0.27%), preterm birth prior to 37 weeks (24.5%; adjusted absolute risk difference, −3.20%; 95% CI, −4.26% to −2.30%), and severe maternal morbidity or death (5.5%; adjusted absolute risk difference, −0.64%; 95% CI, −1.20% to −0.18%).[65]

Cardiovascular Disease

Cardiovascular disease (CVD) is a leading cause of maternal mortality.[17,66] According to Centers for Disease Control and Prevention Pregnancy Mortality Surveillance System data from 665 pregnancy-related deaths in 2024, cardiovascular conditions accounted for 22% of deaths (including cardiomyopathy at 10.7%), and hypertensive disorders of pregnancy caused 7.7% of deaths.[67] Pregnancy-related cardiovascular complications such as hypertension, cerebrovascular disease, and heart failure occur in 15% of US pregnancies and have increased from 11% in 2001 to 13% in 2019 (P < .001).[17]

Box. Commonly Asked Questions About Prepregnancy Care and Counseling

With whom should clinicians discuss reproductive intentions?
Clinicians should discuss reproductive intentions with all reproductive-aged individuals. Questions such as "Would you like to become pregnant in the next year?" initiate a dialogue on reproductive life planning, allowing for individualized counseling and care. For those not planning pregnancy, contraception should be discussed in addition to routine age-specific preventive care.

Use of which substances is associated with increased adverse pregnancy outcomes?
Use of tobacco, alcohol, cannabis, and opioids is associated with adverse pregnancy complications such as preterm birth, low birth weight, neonatal withdrawal, and developmental abnormalities. Brief intervention to increase insights into substance use and education about pregnancy risks can motivate behavioral change. Clinicians should refer individuals identified as at risk of substance use disorder to treatment services.

Which other prepregnancy care and counseling interventions improve outcomes?
Recommended interventions include folic acid supplementation to prevent fetal neural tube defects, immunization for diseases with congenital risks (rubella, varicella, hepatitis B virus), behavioral modification (500- to 1000-kcal/d deficit and ≥150 min/wk of exercise) for individuals with obesity, and optimization of blood glucose (hemoglobin A_{1c} <6.5%) for patients with diabetes. Individuals should be screened before pregnancy for HIV and syphilis, and if infected, should receive treatment with antiretroviral therapy for HIV and benzathine penicillin G for syphilis.

Pregnancy induces hemodynamic adaptations such as expanded blood volume and cardiac output, decreased vascular resistance, and increased heart rate and stroke volume, resulting in increased myocardial workload. Compared with individuals without CVD, those with preexisting CVD have increased peripartum mortality and morbidity, although risk varies substantially across cardiac conditions.[68]

Risk stratification tools, used by clinicians experienced in managing CVD in pregnancy, can complement clinical assessment to guide individualized care. The modified WHO classification, a 4-tier system for estimating maternal risk, ranges from very low risk (class I; eg, small patent ductus arteriosus, mitral valve prolapse) to extremely high risk (class IV; eg, pulmonary arterial hypertension, severe systemic ventricular dysfunction, severe aortic dilation).[69] Other risk stratification tools include CARPREG I,[70] CARPREG II,[71] and Zahara.[72]

The prospective international Registry of Pregnancy and Cardiac Disease has characterized outcomes across a range of CVD, including congenital heart disease, valvular heart disease, cardiomyopathy, and ischemic heart disease. In this registry, spanning 2007 to 2018, maternal mortality, defined as death during pregnancy or within 7 days post partum, occurred in 0.6% of women (34/5739). Heart failure, characterized by events requiring hospital admission, initiation of new treatment, or changes in existing treatment, affected 11% of pregnancies, with the highest frequency in women with pulmonary arterial hypertension, cardiomyopathy, and valvular heart disease.[73] Professional organizations (eg, ACOG, Society for Maternal-Fetal Medicine) recommend that women with CVD receive prepregnancy counseling from a cardiologist and maternal-fetal medicine specialist to evaluate maternal and fetal risks related to pregnancy, assess medications, discuss the potential health effects of pregnancy, and advise about use of contraception if more time is required to optimize prepregnancy health.[12,20,31,73,74]

Multidisciplinary prepregnancy care should address optimization of cardiac status, which may include maximizing medical therapy and discontinuing teratogenic medications (eg, warfarin, angiotensin-converting enzyme inhibitors, statins), surgical intervention (such as valve replacement) if indicated prior to conception, and appropriate diagnostic testing based on a patient's condition and risk category. Testing may include electrocardiography, echocardiography, exercise testing, electrophysiology studies, and genetic evaluation for conditions such as Marfan syndrome, long QT syndrome, and hypertrophic cardiomyopathy.[73] Patients at very high risk of mortality or severe pregnancy complications may be counseled about alternatives to pregnancy, such as gestational carriers or adoption,[12] with consideration of legal, ethical, financial, and psychosocial factors.

Hypertension

Prepregnancy care for individuals with chronic hypertension includes optimizing blood pressure. In contrast to the American College of Cardiology/American Heart Association guideline's target blood pressure of less than 130/80 mm Hg for nonpregnant individuals, ACOG currently recommends a blood pressure of less than 140/90 mm Hg during pregnancy.[75,76] Labetalol and nifedipine are first-line antihypertensive medications during pregnancy.[66] For individuals taking potentially teratogenic antihypertensive medications, changes should be made prior to attempting pregnancy to ensure that blood pressure is well controlled with the new regimen.[77]

Venous Thromboembolism

Pregnancy results in a hypercoagulable state through physiologic alterations in clotting factors, venous stasis due to pelvic vein compression from the gravid uterus, and vascular injury, which can occur during delivery. Thromboembolic events occur in 0.5 to 2 per 1000 pregnancies, and pulmonary embolism accounts for 9.3% of US maternal deaths.[78] Patient-specific risk factors for pregnancy-associated venous thromboembolism include personal history of venous thromboembolism and thrombophilia, such as factor V Leiden or antiphospholipid antibody syndrome. For patients with history of venous thromboembolism or thrombophilia, prepregnancy consultation with a maternal-fetal medicine or hematology specialist can guide counseling of risk, prophylaxis strategies, and medication safety, dosing, and timing.[78,79]

Infectious Diseases

Sexually transmitted infection screening and safe-sex counseling are recommended as part of prepregnancy care.[19,20,80] A meta-analysis of 3 randomized clinical trials (42 200 participants) reported that compared with usual care, prepregnancy behavioral interventions for infection prevention increased condom use (OR, 2.00; 95% CI, 0.70-5.72), reduced sexually transmitted infection prevalence (OR, 0.78; 95% CI, 0.68-0.89), and decreased rates of new sexually transmitted infections and reinfections (OR, 0.65; 95% CI, 0.53-0.80).[81]

Syphilis is a leading cause of adverse pregnancy outcomes including stillbirth, preterm birth, and congenital syphilis, which is associated with sequelae such as jaundice, meningitis, and deformed bones.[18,82] In 2022, 1.1 million pregnant women globally had syphilis, resulting in 700 000 cases of congenital syphilis.[3] In the US, syphilis rates in reproductive-aged women increased from 5.1 per 100 000 in 2017 to 17.7 per 100 000 in 2023.[83] Congenital syphilis rates in the US increased from 50.3 per 100 000 live births in 2019 to 105.8 per 100 000 live births in 2023.[84] Eighty-eight percent of congenital syphilis cases were considered preventable, caused by missed testing or inadequate treatment, and 40% of infants with congenital syphilis were born to individuals who did not receive prenatal care.[85] The highest risk of congenital syphilis occurs during maternal primary and secondary syphilis.[85] Identification and treatment of syphilis with penicillin G benzathine prior to pregnancy can avoid pregnancy complications.

For individuals with HIV infection who are not taking antiretroviral therapy, rates of perinatal HIV transmission range from 15% to 45%, depending on maternal viral load.[86] Maternal HIV infection is associated with neonatal morbidity; a systematic review that included 6 cohort studies, 2 cross-sectional studies, and 1 observational study from 8 countries (Brazil, China, India, Malawi, Nigeria, Tanzania, the US, and Canada) reported that compared with HIV-negative pregnant individuals, neonates born to individuals with HIV had increased odds of preterm birth (OR, 1.57; 95% CI, 1.39-1.78), low birth weight (OR, 1.33; 95% CI, 1.1-1.49), and small for gestational age (OR, 1.38; 95% CI, 1.24-1.53).[87]

Prepregnancy care for individuals with HIV should include initiation or continuation of antiretroviral therapy to attain undetectable viral load, which decreases perinatal HIV transmission rates to approximately 1%.[88,89] Use of antiretroviral therapy has not been associated with fetal anomalies. A 2024 longitudinal US cohort study of 2140 infants exposed to antiretroviral therapy during the first trimester reported similar rates of major congenital anomalies compared with infants not exposed to antiretroviral therapy (darunavir, 6.7% vs 6.6%; aOR, 1.03 [95% CI, 0.62-1.72]; raltegravir, 5.9% vs 6.7%; aOR, 0.91 [95% CI, 0.46-1.81]; rilpivirine, 6.5% vs 6.7%; aOR, 1.04 [95% CI, 0.58-1.85]; elvitegravir, 6.7% vs 6.6%; aOR, 1.31 [95% CI, 0.71-2.41]; dolutegravir, 4.3% vs 6.9%; aOR, 0.76 [95% CI, 0.37-1.57]; and bictegravir, 1.9% vs 6.8%; aOR, 0.34 [95% CI, 0.05-2.51]).[90] Additionally, preexposure prophylaxis should be offered to all sexually active individuals at increased risk of HIV, including those with HIV-positive partners, recent sexually transmitted infections, inconsistent condom use, or frequent sex partners. Sex workers and individuals with injection drug use should also be offered preexposure prophylaxis.[88,91-93]

Substance Use

Pregnancy-associated drug overdose represents a leading cause of maternal death.[32] Substance use disorder contributed to 23.6% of US maternal deaths in 2021, and overdose-related maternal deaths increased from 4.9 per 100 000 to 15.8 per 100 000 from 2018 to 2021.[67,94] In an analysis of 2009 to 2011 Tennessee Pregnancy Risk Assessment Monitoring System data, women who used nonprescribed substances had higher incidence of unintended pregnancy (65.6% vs 48.4%; $P = .003$) and lower contraception use (31.3% vs 46.8%; $P = .02$) compared with those not using these substances.[95] Based on the 2020 National Survey on Drug Use and Health, 8% of US women reported using nonprescribed substances other than alcohol or tobacco during pregnancy.[96]

All individuals of reproductive potential should be screened for tobacco, alcohol, cannabis, opioid, and other substance use with validated tools[19-21,97] such as the TAPS (Tobacco, Alcohol, Prescription, and Other Substance Use) tool.[98] Screening, brief intervention, and referral to treatment (SBIRT) is an approach that identifies substance use and provides early intervention for at-risk individuals.[99] Primary care–based counseling can reduce substance use.[100] Clinicians should provide nonjudgmental advice and education about pregnancy risks and brief interventions including harm reduction strategies such as fentanyl testing strips and referral to needle exchange programs.[101,102]

Tobacco | Tobacco smoke toxins such as nicotine and carbon monoxide impair placental development. In a systematic review and meta-analysis of 142 cohort and case-control studies (total number of patients not available), individuals who smoked during pregnancy had an increased risk of perinatal death (summary RR [sRR], 1.33; 95% CI, 1.25-1.41; 46 studies), stillbirth (sRR, 1.46; 95% CI, 1.38-1.54; 57 studies), and neonatal death (sRR, 1.22; 95% CI, 1.14-1.30; 28 studies) compared with pregnant individuals who did not smoke.[103] Another systematic review and meta-analysis of 55 cohort studies (21 million women) reported increased risk of low birth weight associated with maternal smoking (OR, 1.89; 95% CI, 1.80-1.98).[104] The risk of perinatal mortality rises with increased numbers of cigarettes smoked during pregnancy.[103-105] However, pregnancy often motivates smoking cessation; data from the 2021 US Pregnancy Risk and Assessment Monitoring System (n = 36 493) reported that 56.1% (95% CI, 53.7%-58.5%) of women quit smoking during pregnancy.[106] Smoking cessation is associated with improved pregnancy outcomes. In the abovementioned systematic review and meta-analysis with 142 cohort and case-control studies, former smoking and quitting smoking during pregnancy had similar risks of perinatal death compared with nonsmoking (sRR, 1.02; 95% CI, 0.88-1.19).[103] Clinicians should counsel about risks of smoking on pregnancy and refer individuals to tobacco cessation counseling resources such as quit lines and support groups. Cessation of all nicotine products including e-cigarettes and vaping should also be recommended.[107] ACOG recommends that in addition to psychosocial and behavioral interventions, clinicians should offer medications for smoking cessation (varenicline, bupropion, and nicotine replacement therapy).[107] A retrospective cohort study of 5.2 million births from 2001 to 2020 across 4 countries reported no increased risk of major congenital malformations with prenatal exposure to nicotine replacement therapy (37.6 vs 34.4 per 1000 live births; adjusted RR [aRR], 1.10; 95% CI, 0.98-1.22), varenicline (32.7 vs 36.6 per 1000 live births; aRR, 0.90; 95% CI, 0.73-1.10), or bupropion (35.5 vs 38.8 per 1000 live births; aRR, 0.93; 95% CI, 0.67-1.29) compared with women who did not take these medications.[108]

Alcohol | Consumption of alcohol during pregnancy exposes a developing fetus to ethanol, which passes freely across the placenta. Prenatal alcohol exposure can cause fetal alcohol spectrum disorders, which are a leading cause of intellectual disability and congenital anomalies in the US.[20,109] Because no safe amount of alcohol has been established during any trimester of pregnancy, ACOG recommends total abstinence for women attempting to conceive.[20,99] In a Dutch trial, women receiving prepregnancy care from their general practitioner (n = 211) had reduced alcohol use during the first 3 months of pregnancy compared with women who did not receive primary care (n = 422) (55.7% vs 67.9%; aOR, 1.79; 95% CI, 1.08-2.97).[110]

Cannabis | Cannabis is the most frequently used psychoactive drug during pregnancy; the 2021-2023 National Survey of Drug Use and Health reported a cannabis use prevalence of 6.8% (95% CI, 4.9%-9.3%) among pregnant women.[111-113] A systematic review and meta-analysis (51 studies; 21 146 938 patients) reported that cannabis use during pregnancy was associated with increased odds of low birth weight (OR, 1.75; 95% CI, 1.41-2.18; 20 studies), preterm birth (OR, 1.52; 95% CI, 1.26-1.83; 20 studies), small for gestational age (OR, 1.57; 95% CI, 1.36-1.81; 12 studies), and perinatal mortality (OR, 1.29; 95% CI, 1.07-1.55; 6 studies).[102] During prepregnancy counseling, clinicians should discuss the risk of adverse outcomes associated with cannabis during pregnancy and advise total cessation of cannabis use.[113]

Opioids | In the US, maternal opioid-related diagnoses such as opioid dependence increased from 3.5 per 1000 delivery hospitalizations in 2010 to 8.2 per 1000 delivery hospitalizations in 2017 (absolute difference, 4.6; 95% CI, 3.9-5.4).[114] Neonatal abstinence syndrome, characterized by autonomic and behavioral symptoms such as excessive crying, tachycardia, and tremors, can result from chronic maternal opioid use during pregnancy.[115] In the US between 2010 and 2017, the frequency of neonatal abstinence syndrome increased by 3.3 (95% CI, 2.5-4.1) per 1000 birth hospitalizations—from 4.0 (95% CI, 3.3-4.7) to 7.3 (95% CI, 6.8-7.7) per 1000 birth hospitalizations.[114] Medications for opioid use disorder such as buprenorphine and methadone are safe and effective treatments before and during pregnancy, reducing maternal overdose and death and improving neonatal outcomes.[115-118] Naloxone, a short-acting opioid antagonist, can treat overdose and should be made available to all patients using opioids.[115]

Limitations

This Review has several limitations. First, relevant articles may have been missed. Second, some guideline recommendations are based on low-quality studies and expert consensus. Third, few randomized clinical trials were identified. Fourth, information about reproductive health equity, including racial disparities in pregnancy outcomes and access to care, is not covered.

Conclusions

Prepregnancy counseling and care reduces maternal morbidity and neonatal morbidity and mortality. Primary care–based discussion of reproductive goals, immunizations, screening for infections and substance use, and risk-reducing interventions such as folate supplementation can optimize outcomes in individuals contemplating pregnancy.

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